ModelIt!

Student Activity Pack

Why Do Magnets Work Through Your Phone Case?

Modeling Electromagnetic Fields, Force at a Distance, and Energy Storage

Grade 10L1 | NGSS HS-PS2-3, HS-PS3-5 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How can a magnet attract a paperclip without touching it? What is in the space between them?

2 Why does a magnet stick to a refrigerator door through a piece of paper but not through a thick book?

3 Draw what you think a magnetic field looks like around a bar magnet.

4 Where does the energy come from when a magnet snaps onto metal?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Magnetic Field Strength | Distance from Magnet | Material Type | Force of Attraction | Energy Stored in Field

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Direct Contact

Distance: 0mm | No intervening material

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Through Phone Case

Distance: 3mm | Material: Plastic

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Distance Falloff

Distance: 1mm → 100mm | Material: Air

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Magnetic Domain Alignment — The degree to which atomic magnetic domains in a ferromagnetic material are aligned with the external field — randomly oriented domains produce no net magnetism, while aligned domains create a strong magnetic response

Temperature Effects — Heat disrupts magnetic domain alignment — above the Curie temperature (770°C for iron), ferromagnetic materials lose their magnetism entirely as thermal energy overwhelms magnetic ordering

Electromagnetic Induction — When a magnetic field changes through a conductor, it generates an electric current — this is the principle behind generators, transformers, and wireless charging, connecting magnetism to electricity

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a magnet pull on metal through solid materials — and where does the energy in a magnetic field actually come from? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that magnetic fields are physical entities, not just mathematical descriptions?**

2. **Why does the force drop so dramatically with distance while being unaffected by non-magnetic materials in between?**

3. **What does it mean that the magnetic field stores real energy that can be converted to kinetic energy?**

4. **How would your model change if you placed a ferromagnetic material (iron sheet) between the magnet and the target instead of plastic?**

5. **What are the limitations of modeling electromagnetic interactions with only five components?**

8 STEM Challenge: Design a Magnetic Mounting System

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YOUR ENGINEERING MISSION

Design a magnetic mounting system for a specific application (phone mount, tool holder, or cabinet latch) that provides sufficient holding force while remaining easy to detach when needed.

THE SCENARIO

A product design company wants to create a magnetic phone mount for cars that holds a phone securely during bumps and turns (force requirement: at least 2N) but can be removed with one hand (detach force: less than 8N). The mount must work through any standard phone case up to 5mm thick. Your team must specify the magnet type, size, and arrangement to meet these force requirements.

Engineering Design Thinking

1. WHAT MAGNET MATERIAL AND SIZE WILL PROVIDE THE REQUIRED FIELD STRENGTH?
2. HOW WILL YOU ARRANGE MULTIPLE MAGNETS TO OPTIMIZE THE FORCE PROFILE — STRONG ENOUGH TO HOLD BUT EASY TO SLIDE OFF?
3. WHAT FERROMAGNETIC METAL PLATE SHOULD GO INSIDE THE PHONE CASE, AND HOW THICK MUST IT BE?
4. HOW WILL YOU TEST YOUR PROTOTYPE TO VERIFY IT MEETS THE FORCE REQUIREMENTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Electromagnetic Engineers design systems that use magnetic and electric fields — from MRI machines in hospitals to maglev trains to the motors in electric vehicles. They work in medical device companies, automotive, energy, and defense sectors, earning $75,000–$140,000/year. Materials Scientists who develop new magnetic materials earn $70,000–$130,000/year.